mathoverflow

Is every finite group the outer automorphism group of a finite group?

Asked 5 years ago Modified 1 year, 4 months ago Viewed 3k times

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This question is essentially a reposting of this question from Math.SE, which has a partial answer. YCor suggested I repost it here.

Our starting point is a theorem of Matumoto: every group Q is the outer automorphism group of some group G_Q [1]. It seems to be a research theme to place restrictions on the groups involved. For example, Bumagin and Wise proved that if we restrict Q to be countable then we may take G_Q to be finitely generated [2], and more recently Logan proved that if we restrict Q to be finitely generated and residually finite group then we may take G_Q to be residually finite [3, Corollary D] (this paper also cites quite a few other papers which play this game).

However, all the results I have found always produce infinite groups G_Q, even when the "input" groups Q are finite. For example, Matumoto's groups G_Q are fundamental groups of graphs of groups (so are always infinite), Bumagin and Wise use a variant of Rips' construction (so (as Q is finite) their groups G_Q have finite index in metric small cancellation groups, so are infinite), and Logan's groups G_Q are HNN-extensions of hyperbolic triangle groups (so again are infinite). So we have a question:

> Does every finite group Q occur as the outer automorphism group of some *finite* group G_Q?

The answer is "yes" if we take Q to be finite abelian or a symmetric group; this is what the answer to the original Math.SE question proves.

[1] Matumoto, Takao. "Any group is represented by an outerautomorphism group." *Hiroshima Mathematical Journal* 19.1 (1989): 209-219. (Project Euclid)

[2] Bumagin, Inna, and Daniel T. Wise. "Every group is an outer automorphism group of a finitely generated group." *Journal of Pure and Applied Algebra* 200.1-2 (2005): 137-147. (doi)

[3] Logan, Alan D. "Every group is the outer automorphism group of an HNN-extension of a fixed triangle group." *Advances in Mathematics* 353 (2019): 116-152. (doi, arXiv)

gr.group-theory **finite-groups**

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edited Sep 24, 2020 at 8:49



YCor

asked Sep 24, 2020 at 8:45



user1729

2 As a specific instance, what about $Q = A_5$? – Derek Holt Sep 24, 2020 at 10:42

2 Any finite groups with outer automorphism group isomorphic to Q_8 or A_4? – Mikko Korhonen Sep 24, 2020 at 10:51

2 $C_2 \times C_2 \times C_2 \times C_2$ has (outer) automorphism group isomorphic to A_8, so we can get (at least one) non-abelian simple alternating group. I don't know about A_5. – Carl-Fredrik Nyberg Brodda Sep 24, 2020 at 12:50

1 @KasperAndersen: I don't know. One can show that $\mathrm{Aut}(G) \ncong Q_8$ and $\mathrm{Inn}(G) \ncong Q_8$ for all finite groups G. There is an infinite abelian group G

with $\mathrm{Out}(G) \cong Q_8$, by "de Vries, H.; de Miranda, A. B. Groups with a small number of automorphisms. Math. Z 68 1958 450–464."
– Mikko Korhonen Sep 25, 2020 at 15:24

2 @YCor: I managed to find an Alt_4 example directly: Consider the group (of order $2^7 \cdot 7$) given as the semidirect product of the elementary abelian group $E = (C_2)^6 = \mathbf{F}_8 \times \mathbf{F}_8$ and $Q = C_{14} = C_2 \times C_7$, where Q acts on E by letting C_2 swap the two copies of $\mathbf{F}_8$ and letting $C_7 = \mathbf{F}_8^*$ act diagonally. – Kasper Andersen Sep 30, 2020 at 10:20

3 Answers

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Yes.

For each finite group Q I'll construct a finite group H with $\mathrm{Out}(H) \simeq Q$, moreover H will be constructed as a semidirect product $D \ltimes P$, with P a p-group of exponent p and nilpotency class $< p$, (with prime p arbitrary chosen $> |Q| + 1$) and D abelian of order coprime to p (actually D being a power of a cyclic group of order $p - 1$).

I'll use Lie algebras which are convenient tools to encode p-groups when p is less than the nilpotency class, taking advantage of linear algebra.

In Lie algebras, we denote $[x_1, \dots, x_m] = [x_1, [x_2, \dots, [x_{m-1}, x_m] \cdots]]$. Also I choose the convention to let permutations act on the left.

The base field is $K = \mathbf{F}_p$, p prime. Fix $n \geq 1$. Let $\mathfrak{f}_n$ be the free Lie K-algebra on generators $(e_1, \dots, e_n)$. It admits a unique grading in $\mathbf{Z}^n$ such that e_i has degree E_i, where (E_i) is the canonical basis of $\mathbf{Z}^n$, it is called multi-grading. For instance, $[e_3, [e_1, e_3]]$ has multi-degree $(1, 0, 2, 0, \dots, 0)$.

Let I be a finite-codimensional multi-graded ideal contained in $[\mathfrak{f}_n, \mathfrak{f}_n]$: so the quotient $\mathfrak{g} = \mathfrak{f}_n/I$ is naturally multi-graded. There is a natural action of K^{*n} on $\mathfrak{g}$: namely $(t_1, \dots, t_n)$ acts on $\mathfrak{g}_{(m_1,\dots,m_n)}$ by multiplication by $\prod_{i=1}^n t_i^{m_i}$. Let $D \subset \mathrm{Aut}(\mathfrak{g})$ be the image of this action. Also denote by c the nilpotency class of $\mathfrak{g}$: we assume $p > c$.

Using that $p > c$, we endow, à la Malcev–Lazard, $\mathfrak{g}$ with the group law given by the Baker-Campbell-Hausdorff formula: $xy = x + y + \frac{1}{2}[x, y] + \dots$. We thus view $\mathfrak{g}$ as both a Lie algebra and a group; we denote it as G when endowed with the group law (but feel free to refer to the Lie algebra law in G); this is a p-group of exponent p and nilpotency class $c < p$. Define $H = D \ltimes G$.

Every permutation $\sigma \in \mathfrak{S}_n$ induces an automorphism u_σ of $\mathfrak{f}_n$, defined by $u_\sigma(e_i) = e_{\sigma(i)}$. Write $\Gamma_I = \Gamma_{\mathfrak{g}} = \{\sigma \in \mathfrak{S}_n : u_\sigma(I) = I\}$.

Proposition 1. The natural map $\Gamma_{\mathfrak{g}} \to \mathrm{Out}(H)$ is an isomorphism.

We need a lemma:

Lemma 2. Define M as $\mathbf{F}_p^* \ltimes \mathbf{F}_p$. Then $\mathrm{Out}(M^n)$ is reduced to the symmetric group permuting the n factors.

Proof. Let f be an automorphism. This is a product of n directly indecomposable center-free groups, hence its automorphism group permutes the n (isomorphic) factors. Hence after composing with a permutation, we can suppose that f preserves each factor. We are then reduced to checking that every automorphism of $\mathbf{F}_p^* \ltimes \mathbf{F}_p$ is inner. Indeed after composing by an inner automorphism, we can suppose that it maps the Hall subgroup $\mathbf{F}_p^*$ into itself. Then after composing with an inner automorphism in the given copy of $\mathbf{F}_p^*$, we can also suppose it acts as identity on $\mathbf{F}_p$. It easily follows that this is the identity. (Note for conciseness I used some slightly fancy results in this proof, but this lemma can be checked more elementarily.) □

Lemma 2'. [added May 8, 2024] Let f be an automorphism of $H = D \ltimes G$ preserving D such that the automorphism of $H/[G,G] \simeq M^n$ induced by f is a plain permutation σ of the n factors. Then $\sigma \in \Gamma_I$ and f permutes the e_i as σ.

Proof. Write $f(e_i) = e_{\sigma(i)} + c_i$, where $c_i \in [\mathfrak{g}, \mathfrak{g}]$. Write the graded decomposition $c_i = \sum_{k \in W_n} c_{i,k}$, where W_n is the set of n-tuples of non-negative integers of sum ≥ 2. For every $d \in D$ and every i, we have $f(de_id^{-1}) = f(d)f(e_i)f(d)^{-1}$. Writing $d \cdot g$ for dgd^{-1}, this rewrites successively

$$f(d \cdot e_i) = f(d) \cdot f(e_i)$$

(writing $d = (d_1, \ldots, d_n)$): for $d_1, \ldots, d_n \in \mathbf{F}_p^*$

$$f(d_i e_i) = (d_{\sigma^{-1}(1)}, \ldots) \cdot (e_{\sigma(i)} + \sum_{k \in \mathbf{Z}^n} c_{i,k}.$$

$$d_i f(e_i) = d_i e_{\sigma(i)} + \sum_{k \in \mathbf{Z}^n} d_1^{k_{\sigma^{-1}(1)}} \ldots d_n^{k_{\sigma^{-1}(n)}} c_{i,k}.$$

Taking the component of degree k we get (for every k, d, i)

$$d_i c_{i,k} = d_1^{k_{\sigma^{-1}(1)}} \ldots d_n^{k_{\sigma^{-1}(n)}} c_{i,k}.$$

Thus, for every i, k such that $c_{i,k} \neq 0$, we obtain $d_i = d_1^{k_{\sigma^{-1}(1)}} \ldots d_n^{k_{\sigma^{-1}(n)}}$ for all $d_1, \ldots, d_n \in \mathbf{F}_p^*$. So, for such i, k, $d_i^{k_{\sigma^{-1}(i)}} - 1) = 1$ and $d_j^{k_{\sigma^{-1}(j)}} = 1$ for all $d_1, \ldots, d_n$. This means that, for each such (i,k) $p-1$ divides $k_{\sigma^{-1}(i)} - 1$ and divides $k_{\sigma^{-1}(j)}$ for all $j \neq i$. Since $c_{i,k} = 0$ whenever component of k is $\geq p-1$ (because the nilpotency length is $< p$ and each commutator involves at least two generators), this means that for every i, k such that $c_{i,k} \neq 0$, we have $k_i = 1$ and $k_j = 0$ for all $j \neq i$. But since k has total sum ≥ 2, this is impossible. We deduce that $c = 0$, i.e., f permutes the generators.

Proof of the proposition. Let f be an automorphism of $H : D \ltimes G$. After composing with an inner automorphism, we can suppose that f maps the Hall subgroup D into itself.

Next, f induces an automorphism of $H/[G,G]$, which can naturally be identified with M^n of the previous lemma (recall that $I \subset [G,G] = [\mathfrak{g},\mathfrak{g}]$). Hence after composing with conjugation by some element of D, we can suppose that f both preserves D and acts on $H/[G,G] \simeq M^n$ by permuting the factors (without changing coordinates). **[Edit May 8, 2024]** By Lemma 2', f permutes the generators e_i and is induced by an element of Γ_I.**[End edit** — a few lines of the preceding proof now erased]$\square$.

To conclude we have to prove:

Proposition 3. For every finite group Q of order n and prime $p > n+1$ there exist n, p and I finite-codimensional multi-graded ideal in $\mathfrak{f}_n (= \mathfrak{f}_n(\mathbf{F}_p)$, such that $\Gamma_I \simeq Q$. (And such that $\mathfrak{f}_n/I$ has nilpotency class $\leq n+1$.)

This starts with the following, which provides for each finite group Q a relation whose automorphism group is the group $L_Q \subset \mathfrak{S}(Q)$ of left translations of Q.

Lemma 4. Let Q be a group, $n = |Q|$, and $q = (q_1, \dots, q_n)$ an injective n-tuple of Q. Define $X = Qq \subset Q^n$. Then $L_Q = \{\sigma \in \mathfrak{S}(Q) : \sigma X = X\}$.

Proof. Clearly L_Q preserves X. Conversely, if σ preserves X, after composing with a left translation we can suppose that $\sigma(q_1) = q_1$, so $\sigma(q) \in \{q_1\} \times Q^{n-1}$; since $X \cap \{q_1\} \times Q^{n-1} = \{q\}$, we deduce $\sigma(q) = q$, which in turn implies $\sigma = \mathrm{Id}$. $\square$.

Proof of the proposition. Write $\mathfrak{f}_Q \simeq \mathfrak{f}_n$ as the free Lie algebra over the generating family $(e_q)_{q \in Q}$. It can be viewed as graded in $\mathbf{Z}^G$, with basis $(E_q)_{q \in Q}$. Write $E = \sum_q E_q$.

For $q \in Q^n$, define $\xi_q = [e_{q_n}, e_{q_1}, e_{q_2}, \ldots, e_{q_n}]$ (note that it is homogeneous of degree $E + E_{q_n}$; in particular $(\xi_h)_{h \in X}$ is linearly independent. Fix an injective n-tuple q and define X as in the proof of the lemma; for convenience suppose $q_n = 1$. Define J as the n-dimensional subspace of $\mathfrak{f}_Q$ with basis $(\xi_h)_{h \in X}$.

Define $I = J \oplus \mathfrak{f}_Q^{n+2}$, where $\mathfrak{f}_Q^i$ is the i-th term in the lower central series. Hence I is an ideal, and $\mathfrak{g} = \mathfrak{f}_Q / I$ is defined by killing all i-fold commutators for $i \geq n+1$ and certain particular $(n+1)$-fold commutators. (Since we assume $p > n+1$, we can view it as a group as previously.)

Claim. For $h = (h_1, \ldots, h_n) \in Q^n$ with $h_{n-1} \neq h_n$, we have $\xi_h \in I$ if and only if $h \in X$.

By definition, $h \in X$ implies the condition. Now suppose that h satisfies the condition. First, the condition $h_{n-1} \neq h_n$ ensures that $\xi_h \neq 0$; it is homogeneous in the multi-grading. If it belongs to J, its multi-degree is therefore some permute of $(2, 1, \ldots, 1)$. This is the case if and only if the h_i are pairwise distinct, so we now assume it; its degree is therefore equal to $E_{h_n} + E$. Now $J_{E + E_{h_n}}$ is 1-dimensional, and generated by $\xi_{h_n q}$. Hence ξ_h is a scalar multiple of $\xi_{h_n q}$:

$$[e_{h_n}, e_{h_1}, \ldots, e_{h_{n-1}}, e_{h_n}] = \lambda [e_{h_n}, e_{h_n q_1}, \ldots, e_{h_n q_{n-1}}, e_{h_n}].$$

The next lemma implies that $h_i = h_n q_i$ for all $i \in \{1, \ldots, n-1\}$. So $h \in X$. The claim is proved.

The claim implies that for every permutation σ of Q, if the automorphism u_σ of $\mathfrak{f}_Q$ preserves I, then σ has to preserve X, and hence (Lemma 4) σ is a left translation of Q. This finishes the proof. $\square$.

Lemma 5 Consider the free Lie algebra on $(e_1, \ldots, e_n)$. If for some permutation σ of $\{1, \ldots, n-1\}$ and scalar λ we have

$$[e_n, e_1, \dots, e_{n-1}, e_n] = \lambda [e_n, e_{\sigma(1)}, \dots, e_{\sigma(n-1)}, e_n],$$

then σ is the identity and $\lambda = 1$.

Proof. Use the representation f in $\mathfrak{gl}_n$ mapping e_i to the elementary matrix $\mathcal{E}_{i-1,i}$ (consider indices modulo n). Then $[e_n, e_1, \dots, e_{n-1}, e_n] = [e_n, e_1, [e_2, \dots, e_n]]$ maps to

$$[\mathcal{E}_{n-1,n}, \mathcal{E}_{n,1}, \mathcal{E}_{1,n}] = [\mathcal{E}_{n-1,n}, \mathcal{E}_{n,n} -, \mathcal{E}_{1,1}] = \mathcal{E}_{n-1,n}.$$

Let by contradiction j be maximal such that $\sigma(j) \neq j$; note that $2 \leq j \leq n-1$ and $n \geq 3$. Then $[e_n, e_{\sigma(1)}, \dots, e_{\sigma(n-1)}, e_n] = [e_n, e_{\sigma(1)}, \dots, e_{\sigma(j)}, [e_{j+1}, \dots, e_n]]$ maps to

$$w = [\mathcal{E}_{n,n-1}, \mathcal{E}_{\sigma(1)-1,\sigma(1)}, \dots, \mathcal{E}_{\sigma(j)-1,\sigma(j)}, \mathcal{E}_{j,n}],$$

which cannot be zero. Hence $[\mathcal{E}_{\sigma(j)-1,\sigma(j)}, \mathcal{E}_{j,n}] \neq 0$. Since $\sigma(j) < j$, this implies $\sigma(j) - 1 = n$ (modulo n), that is, $\sigma(j) = 1$. So

$$w = -[\mathcal{E}_{n,n-1}, \mathcal{E}_{\sigma(1)-1,\sigma(1)}, \dots, \mathcal{E}_{\sigma(j-1)-1,\sigma(j-1)}, \mathcal{E}_{j,1}].$$

In turn, $[\mathcal{E}_{\sigma(j-1)-1,\sigma(j-1)}, \mathcal{E}_{j,1}]$, using that $\sigma(j-1) \neq 1$, implies $\sigma(j-1) = j$, and so on, we deduce that σ is the cycle $j \mapsto j+1$ (modulo $n-1$). Eventually we obtain

$$w = [\mathcal{E}_{n-1,1}, \mathcal{E}_{1,2}, \mathcal{E}_{2,1}] = \mathcal{E}_{n-1,1}.$$

So $\mathcal{E}_{n-1,n} = \lambda \mathcal{E}_{n-1,1}$, contradiction.

(Note on Lemma 5: one has $[e_1, e_2, e_1, e_2] = [e_2, e_1, e_1, e_2]$, but this and its obvious consequences are probably the only identities between Lie monomials in the free Lie algebra, beyond the ones obtained from skew-symmetry in the last two variables.)

Note on the result: the resulting group H roughly has size $|Q|^{|Q|}$, which is probably not optimal.

In Proposition 4, I is strictly contained in $\mathfrak{f}_Q^{n+1}$, as soon as $|Q| \geq 3$, so the nilpotency class of G is then equal to $n+1$. (For $|Q| = 1$, choosing $p \geq 3$ outputs H as the group $M = M_p$ which has trivial Out, and G is abelian then; for $|Q| = 2$, this outputs a group H of order $(p-1)^2.p^3$ for chosen prime $p \geq 5$, and G has nilpotency length 2. For $|Q| = 3$ it outputs a group H of order $(p-1)^3.p^{29}$ for $p \geq 5$ which is already quite big.) To improve the bounds in explicit cases by running this method, one should describe Q a permutation group of a set Y that can be described as stabilizer of some ℓ-ary relation R contained in the set of pairwise distinct ℓ-tuples of Y, for ℓ as small as possible.

Share Cite Improve this answer Follow edited May 8, 2024 at 19:25

answered Sep 25, 2020 at 11:50



YCor

13 Have you turned this into a paper? (it's worth it) – Brauer Suzuki Oct 9, 2021 at 11:48

In the proof of Prop1: f permutes the factors of $H/[G,G]$, but in the subsequent sentence f is already the identity on D. How is this possible? Your proof of Lemma 5 works inside a non-nilpotent Lie algebra. Can the result actually be used for the nilpotent algebra at hand? – Brauer Suzuki May 8, 2024 at 17:13

@BrauerSuzuki You're right. I've fixed the proof, with an additional lemma (and then some simplification). It consists in directly checking that in the given group H, being a permutation of factors modulo $[G,G]$ implies permuting the given generators of G. – YCor May 8, 2024 at 19:27

Thanks. One more thing: I think the order of magnitude of $|H|$ is about $|Q|^{|Q|^{|Q|}}$ (explained in my paper). – Brauer Suzuki May 9, 2024 at 4:42

14

In [Bryant, R. M.; Kovács, L. G., Lie representations and groups of prime power order. J. London Math. Soc. (2) 17 (1978), no. 415-421], the authors come close to proving what you ask: they show that any group $G < \mathrm{GL}(V)$, with V an elementary abelian p-group, can be realized as the image of $\mathrm{Aut}(P) \to \mathrm{Aut}(P/\Phi(P)) = \mathrm{GL}(V)$, for some finite p-group P with Frattini quotient $P/\Phi(P) = V$.

I used this in a paper with Frank Adams in the late 1980s, and we also reference Theorem 13.5 of Huppert and Blackburn, *Finite Groups II*. Section 13 of that book is all about automorphisms of p-groups. My memory (30 years old) is that exactly the result you are interested in was proved there. (This is not freely available online, so I can't easily check this right now.)

Added a little bit later: I should perhaps have explicitly pointed out that the map $\mathrm{Aut}(P) \to \mathrm{Aut}(P/\Phi(P))$ factors through $\mathrm{Aut}(P) \to \mathrm{Out}(P)$. Also the kernel of this map is a p-group: see e.g. [(24.1), Aschbacher, *Finite Group Theory*]. So one knows that, for any prime p, any finite group is the quotient of $\mathrm{Out}(P)$ for some finite p-group P, with another p-group as kernel.

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edited May 8, 2024 at 17:41

LSpice

answered Sep 24, 2020 at 13:36

Nicholas Kuhn

A quote from the remark at the end of §13 of the above reference: "Theorem 13.5 shows that any subgroup of $GL(n,p)$ is the linear group induced on $\mathfrak{P}/\Phi(\mathfrak{P})$ by Aut$(\mathfrak{P})$ for some p-group $\mathfrak{P}$. Since any finite group is isomorphic to a subgroup of $GL(n,p)$ for some n, every finite group is isomorphic to the group induced on $\mathfrak{P}/\Phi(\mathfrak{P})$ by Aut$(\mathfrak{P})$ for some p-group $\mathfrak{P}$." – Carl-Fredrik Nyberg Brodda Sep 24, 2020 at 13:50

(The book is available online, albeit from canonical non-MO-linkable places). – Carl-Fredrik Nyberg Brodda Sep 24, 2020 at 13:52

@Carl-FredrikNybergBrodda Thanks for looking this up. So I guess they basically say the same as Bryant and Kovacs. – Nicholas Kuhn Sep 24, 2020 at 15:02

2 Also the induced map $\mathrm{Out}(P) \to \mathrm{Aut}(P/\Phi P)$ has its kernel a p-group. Hence this shows that for every finite group G and prime p there exists a finite p-group H and a surjective map $\mathrm{Out}(H) \to G$ whose kernel is a p-group. – YCor Sep 24, 2020 at 15:07

@YCor Thanks for pointing out this classic fact. I've added this into my answer. – Nicholas Kuhn Sep 24, 2020 at 15:27

5

I have allowed myself to translate @YCor's proof into the language of group theory. Doing so, I clarified some (probably minor) issues. The result is given in the second half of this arxiv paper: https://arxiv.org/abs/2405.02992

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edited May 8, 2024 at 18:53



Emil Jeřábek

answered May 8, 2024 at 17:09

B

Brauer Suzuki

7 Although it's not my place to argue, I find myself uncomfortable with the idea of taking someone else's MO answer and posting a version of it on the arXiv. But perhaps you communicated separately with @YCor before doing so. – LSpice May 8, 2024 at 17:45

5 We didn't talk separately. – YCor May 8, 2024 at 18:41

5 @LSpice: Proper credit has been given, though, together with a link to the YCor's MO answer. Furthermore, the proof published on arXiv diverges from the one on MO significantly, by replacing the Lie group-theoretical arguments with purely group-theoretical ones. Therefore, it is not a reposting of YCor's proof, but a modification of it. – Alex M. May 8, 2024 at 18:55

8 I'll split the difference between LSpice and Alex M. and say that while the article gives appropriate attribution, it still would have been best to reach

out to YCor *before* posting to the arXiv. – Sam Hopkins ♦ May 8, 2024 at 19:42

2 @SamHopkins You can see under his answer that I asked YCor already in 2021 about a paper. I never got a reply. – Brauer Suzuki May 9, 2024 at 4:39

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